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inedemogR: Reproducible retrieval and life-table computation for Spanish subnational demographic data

J.R. Caro-Barrera

Universidad de Córdoba, Córdoba, Spain

HIGHLIGHTS

- `inedemogR` builds Spanish province life tables from INE's public API.
- It retrieves and harmonises INE births, deaths and population with per-stage checks.
- Life tables follow HMD Protocol V6; adaptations to INE's data are stated explicitly.
- Provincial e_0 and e_{65} agree with INE's published tables in about 99% of cases.
- Frozen inputs and scripts regenerate every reported number and figure.

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ABSTRACT

`inedemogR` is an R package that turns the Spanish National Statistics Institute (INE) demographic data into tidy, analysis-ready tables. A quick-lookup interface returns population, birth and death totals by geography and year. A second interface builds province-level exposures, death rates and single-year period life tables following the Human Mortality Database protocol (Version 6), adapted where INE's data require it: deaths are split evenly between Lexis triangles and old-age mortality is fitted at ages 80–99. Single-year life tables are supported from 2002. In 3588 province–sex–year comparisons for 2002–2024, made separately for life expectancy at birth and at age 65, the package agrees with INE's published provincial tables to within 0.25 years in approximately 99% of cases.

Table 1
Code metadata.

Nr.	Code metadata description	Metadata
C1	Current code version	v0.2.1
C2	Permanent link to code/repository used for this code version	https://github.com/jrcarob/inedemogR_package (commit `76e429e`); release 0.2.1 on CRAN [15]
C3	Legal Code License	MIT License
C4	Code versioning system used	git
C5	Software code languages, tools, and services used	R ($\geq$ 4.1.0)
C6	Compilation requirements, operating environments & dependencies	R packages: dplyr, ggplot2, httr2, ineapir, mapSpain, purrr, readr, rlang, scales, sf, stringi, stringr, tibble, tidyr. Live internet access to the INE dissemination API required for data retrieval.
C7	If available, link to developer documentation/manual	Package vignette `vignette("inedemogR-tutorial")` and reference manual, both shipped with the package; validation study in `reproducibility/`
C8	Support email for questions	jrcaro@uco.es

Email address: z52cabaj@uco.es.

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1. Motivation and significance

1.1. The gap between INE's API and analysis-ready demographic data

The Spanish National Statistics Institute (INE) publishes population figures, vital statistics and life tables through a public dissemination API that requires no registration. The R wrapper `ineapir` [1] gives general access to that API, including multidimensional filters, flattening and metadata. What it returns is INE's own table structure: Spanish category labels, totals mixed with their breakdowns, small-count suppression, and different geographic levels for different series. Turning these tables into demographic inputs (complete age-by-sex grids per province and year, with structural zeros told apart from missing values) is work that every analyst repeats and few publish. It is also easy to get wrong (Table 1).

Other packages address neighbouring needs. `mapSpain` [2] supplies Spanish administrative geometries, `ineAtlas` [3] covers INE's household income atlas, and `tidycensus` [4] plays this role for the United States census data. `inedemogR` provides a demography-specific, tidy layer over INE's population, birth and death series.

1.2. From the API to a subnational life table

Subnational life tables are central to the study of regional mortality inequalities. INE publishes provincial life tables computed from death microdata with its own methodology [13], but the series is fixed: users cannot vary the method, extend the tables, or inspect intermediate quantities. The Human Mortality Database (HMD) protocol [5] is the international reference for life tables computed from aggregate data. To our knowledge, no published software combines retrieval of INE's public aggregate province data with automated computation of HMD-protocol provincial life tables. Table 2 sets the closest packages side by side.

The analytical functions of `MortalityLaws`, `demography` or `MortCast` can build provincial life tables once a user has assembled provincial deaths and exposures. They do not assemble them from INE's data, and that assembly is what `inedemogR` adds. Its demographic methods are established. The package runs retrieval, harmonisation, validation and the HMD procedures as one documented sequence, with every intermediate table available.

1.3. Main contribution

`inedemogR` automates the path from INE's public API to province-level period life tables: retrieval of age- and sex-specific province births, deaths and population; exposure-to-risk; 1×1 and 5×1 central death rates; single-year life tables following HMD Protocol Version 6, with the adaptations INE's data require are stated explicitly (Section 2.3); and fertility, decomposition and summary indicators on the same objects. Compared with assembling the inputs by hand, the cleaning rules (age recoding, small-cell recovery, province matching, coverage checks) are written once, tested and versioned, and every intermediate quantity can be inspected and recomputed from frozen inputs (Section 2.4). The package can be used as a data layer only, or as an analysis toolkit on data obtained elsewhere in the documented tidy shape.

Table 2
Data access and analytical scope of related R packages.

Package	Data access	Analytical scope for Spanish provinces
`ineapir` [1]	INE API, any table	None (retrieval only)
`qlifetable` [6]	User-supplied microdata	Quarterly life tables; needs restricted-access records
`MortalityLaws` [10]	HMD download; user data	Life tables and old-age laws (incl. Kannisto) from user-supplied rates or deaths and exposures
`demography`, `MortCast` [7,8]	User data	Generic life-table, smoothing and projection tools
`HMDHFDplus` [9]	HMD/HFD download	HMD holds national Spanish series only
`inedemogR`	INE API (aggregate)	Retrieval, harmonisation, exposure, rates, HMD-protocol life tables, decomposition, fertility indicators

2. Software description

`inedemogR` has two retrieval front ends that share a tidying, validation, caching and geometry layer (Fig. 1).

2.1. Software architecture

The *quick-lookup* front end (`get_ine_demog`, `get_ine_geo`, `plot_ine_map`, `list_ine_indicators`, `update_ine_data`) returns population, birth and death totals, one tidy row per geography and year, through `ineapir`, with optional `mapSpain` geometry. An indicator registry (`ine_variables`) maps readable names to INE table identifiers.

The *mortality-workflow* front end (`get_ine_population`, `get_ine_deaths`, `get_ine_births`, `get_ine_births_by_age`) queries INE's Tempus3 service directly with `httr2`. The province × age × sex population table exceeds the JSON service's volume limit, so the package falls back to INE's bulk CSV export and parses both formats into a single shape. The output feeds a fixed sequence of pure functions (`compute_exposure`, `compute_death_rates`, `build_life_tables`), each consuming and returning a tidy data frame, so any step can be inspected, replaced or tested on its own. `download_ine_data` writes any part of the workflow to CSV or fixed-width HMD-format files.

2.2. Software functionalities

Retrieval and tidying. Long tibbles with recoded age and sex, parsed geographic codes, totals separated from components, and a provenance record (Section 2.4).

Mortality. Exposure-to-risk; 1×1 and 5×1 central death rates and age-standardised rates (2013 European Standard Population); single-year and abridged period life tables; and decomposition of a life-expectancy difference by age (Arriaga or Pollard).

Indicators. Age dependency, ageing index and sex ratio; crude birth and death rates, general fertility, infant mortality and natural increase; age-specific and total fertility, mean age at childbearing, and gross and net reproduction rates; e_0 and e_{65} summaries.

Visualisation. Population pyramids, time series, Lexis surfaces and choropleth maps.

2.3. Implemented methods

Exposure and rates. Following HMD Protocol V6, Eq. 57 [5],

$$E(x,t) = \tfrac{1}{2}\big[P(x,t) + P(x,t+1)\big] + \tfrac{1}{6}\big[D^L(x,t) - D^U(x,t)\big], \tag{1}$$

where P are January 1 stocks and D^L, D^U are deaths in the lower (cohort born in $t-x$) and upper Lexis triangles. INE's death tables are not classified by cohort, so deaths are split evenly and exposure is the mean of the two stocks. Rates are $m(x,t) = D/E$; grouped rates pool deaths and exposures; the age-standardised rate uses the 2013 European Standard Population [16], whose open group 95+ (weight 200 of 100,000) is spread over ages 95–99 and 100+.

Life tables. The single-year tables follow HMD Protocol V6, Section 7.1. For a_0 they use the sex-specific Andreev–Kingkade formulas [11] (Table 3 of the HMD Protocol), e.g., $a_0 = 0.14903 - 2.05527\, m_0$ for females with $m_0 < 0.01724$, and $a_x = \frac{1}{2}$ at all other ages. Old-age rates come from the Kannisto model [12],

$$\mu_x = \frac{a\, e^{b(x-80)}}{1 + a\, e^{b(x-80)}}, \quad a, b \geq 0, \qquad D_x \sim \text{Poisson}\big(E_x\, \mu_{x+1/2}\big), \tag{2}$$

fitted by maximum likelihood and used from age Y upward, where Y is the lowest age from 80 with at most 100 female or 100 male deaths, $80 \leq Y \leq 95$. Then

$$q_x = \frac{m_x}{1 + (1 - a_x)\, m_x}, \qquad L_x = l_x - (1 - a_x)\, d_x, \qquad e_x = \frac{1}{l_x} \sum_{y \geq x} L_y, \tag{3}$$

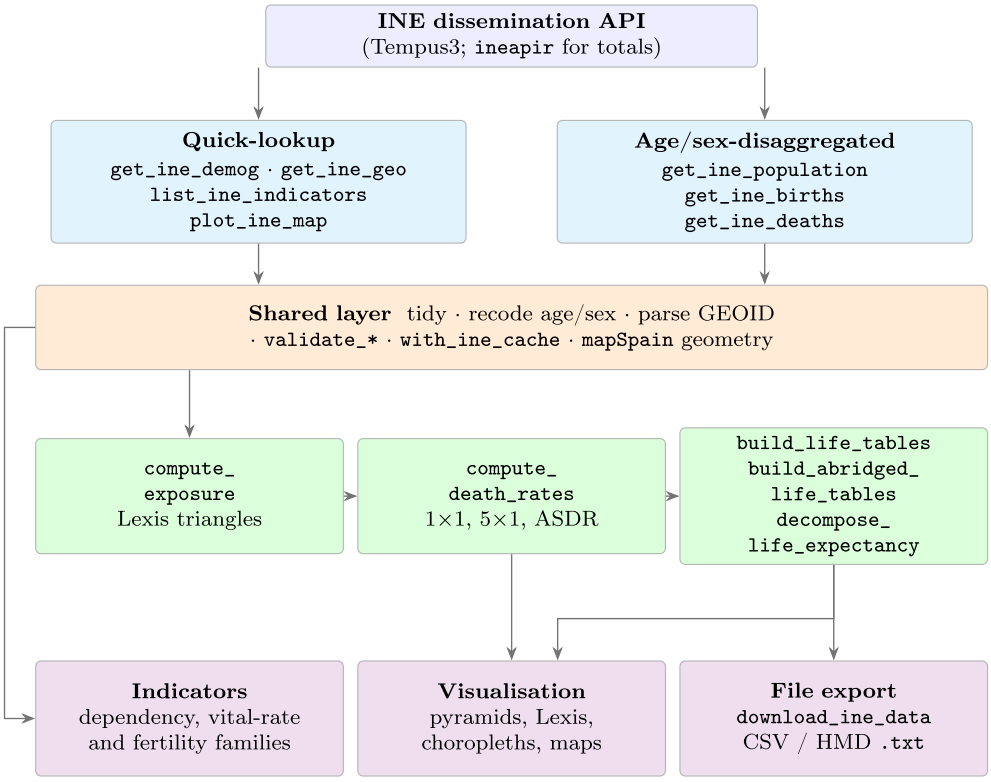


Fig. 1. Architecture of `inedemogR`. Two retrieval front ends feed a shared tidy/validate/cache/geometry layer. The mortality workflow is a fixed sequence of pure functions, each consuming and returning a tidy data frame; its output, together with the shared layer's, feeds the indicator, visualisation, and file-export functions.

with radix 10^5 and the open interval at 110+ ($L = l/m$). Combined-sex tables use the death-weighted a_0 and smoothed exposure weights at ages $\geq Y$ (Protocol Eqs. 69–77). *Adaptations* forced by INE's data are: the even triangle split (the Protocol uses a regression split and a birth-month correction), and a Kannisto fit on ages 80–99 instead of 80–110+, because INE's single-year detail ends at 100+; the female share of exposure is smoothed on the same ages. Their effect is quantified in Section 3.3. Abridged tables are a separate estimate from grouped observed rates, without smoothing; they use Greville's ${}_na_x$ with

$$ {}_nq_x = \frac{n\,{}_nm_x}{1 + (n - {}_na_x)\,{}_nm_x}, \qquad {}_nL_x = n\,l_{x+n} + {}_na_x\,{}_nd_x, \tag{4} $$

so ${}_nm_x = {}_nd_x/{}_nL_x$ holds exactly [17].

Decomposition. Arriaga's method [18] is exact: the contributions sum to $e_0^{(2)} - e_0^{(1)}$. Pollard's symmetric form [19],

$$ \Delta(x) = \tfrac{1}{2}\left(m_x^{(1)} - m_x^{(2)}\right)\left(\frac{L_x^{(2)}}{l_0}\,\bar{e}_x^{(1)} + \frac{L_x^{(1)}}{l_0}\,\bar{e}_x^{(2)}\right), \tag{5} $$

with $\bar{e}_x = \frac{1}{2}(e_x + e_{x+1})$, integrates each closed year by the midpoint rule and the open interval exactly,

$$ \Delta(\omega) = \left(m_\omega^{(1)} - m_\omega^{(2)}\right)\, e_\omega^{(1)} e_\omega^{(2)}\, \frac{l_\omega^{(1)} + l_\omega^{(2)}}{2\,l_0}. \tag{6} $$

Both methods return the residual between the summed contributions and the actual difference.

Fertility. Following INE's methodology [14], $\mathrm{ASFR}(x) = B(x)/\bar{W}(x)$ in births per woman, where $\bar{W}(x)$ is the mean of the January 1 female stocks of t and $t+1$, and births to mothers under 15 or over 49 are added to ages 15 and 49. Then

$$ \mathrm{TFR} = \sum_x \mathrm{ASFR}(x), \qquad \mathrm{MAC} = \frac{\sum_x \left(x + \frac{1}{2}\right)\mathrm{ASFR}(x)}{\sum_x \mathrm{ASFR}(x)}, $$
$$ \mathrm{GRR} = \sum_x \mathrm{ASFR_f}(x), \qquad \mathrm{NRR} = \sum_x \mathrm{ASFR_f}(x)\,\frac{L_x}{l_0}. \tag{7} $$

2.4. Missing data, failure behaviour and reproducibility

INE omits cells with no deaths, and the package treats an absent age within a reported province-year as a structural zero. A sex-specific count suppressed by INE is recovered as the total minus the other sex when both are published. Otherwise it stays missing, and the missing value propagates to the rate, the age-standardised rate and the validation result. A province-year with no death records is excluded, never set to zero mortality. Life tables are built only when exposure is available at every age 0–100 and rates are finite below Y; other province-years are withheld and listed with the reason. INE's provincial population is top-coded at 85+ before 2002, so the supported period starts in 2002. Validators fail, rather than warn, on missing ages, non-finite values, a broken $m_x = d_x/L_x$ identity or an implausible e_0. Abridged tables are built only from unique, contiguous age groups from 0–4 to a single open interval, and when deaths are supplied by birth cohort a missing triangle count leaves that exposure missing.

A fixed package version and a live API do not by themselves guarantee identical results, because INE revises its series. Every retrieved object therefore carries a provenance record (request, retrieval time, package version). Results are cached on disk in a user-chosen directory, or for the session if none is given, and are refreshed only on request. If the

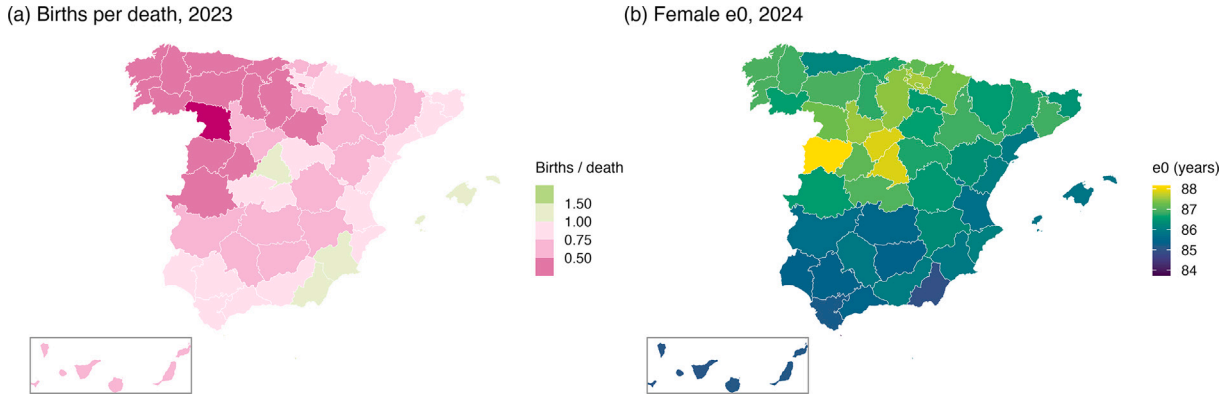


Fig. 2. (a) Births per death by province, 2023 (`map_indicator`); (b) Female life expectancy at birth, 2024 (`map_life_expectancy`). Almost every mainland province records more deaths than births; (b) Shows the north–south e_0 gradient.

API is unavailable or a response cannot be parsed, retrieval stops with an error. The repository's `reproducibility/` folder holds the frozen inputs, the scripts that regenerate every reported result, and a record of the R session. Version 0.2.1 is released on CRAN [15], which keeps every published version permanently in its archive.

2.5. Testing and validation

Three kinds of evidence are kept separate. (i) *Data-integrity checks* (`validate_*`) run on each retrieved or computed table. (ii) *Software tests*: version 0.2.1 has 126 `testthat` tests with fixed inputs, including the review's controlled cases (missing counts, missing ages, missing or duplicated age groups, terminal-only mortality change, constant mortality under standardisation) and identity checks ($\sum d_x = l_0$, $m_x = d_x/L_x$, exact Arriaga additivity, antisymmetric decompositions). Line coverage is 68% overall and 90–100% for the computational modules. An opt-in test runs the full workflow on live INE data; it runs weekly in continuous integration, separately from `R CMD check`, which passes on Linux (R release, devel and previous release), macOS and Windows. Coverage is reported to show the scope of the tests and says nothing about numerical accuracy, which (iii) addresses. (iii) *Numerical validation* against INE's published life tables and fertility indicators, with sensitivity analyses (Sections 3.3 and 3.5).

3. Illustrative examples

All results below come from the frozen inputs and the scripts in `reproducibility/`. Each example also corresponds to a script in `examples/`, and `vignette("inedemogR-tutorial")` walks through every function.

3.1. Quick lookup: births per death

```
library(inedemogR)
vitals <- get_ine_demog(indicator = c("births_total", "deaths_total"),
                        year = 2023)
map_indicator(birth_death_ratio(vitals), value_col = "birth_death_ratio",
              geo_level = "province", palette = "PiYG",
              breaks = c(0.25, 0.5, 0.75, 1, 1.5, 2, 3))
```

The resulting map is Fig. 2(a).

3.2. Mortality workflow: life expectancy by province

```
cache    <- tools::R_user_dir("inedemogR", "cache")
pop      <- get_ine_population(n_periods = 100, cache_dir = cache)
deaths   <- get_ine_deaths(n_periods = 100, cache_dir = cache)
exposure <- compute_exposure(pop$data, deaths$data_provinces)
rates    <- compute_death_rates(deaths$data_provinces, exposure$data)
lt       <- build_life_tables(rates$mx_1x1)  # lt$failed: withheld years
le       <- life_expectancy_summary(lt$fltper, sex = "female")
map_life_expectancy(le, year = 2024, sex = "female")
```

Table 3
Female e_0 and e_{65} (years), 2024, from `life_expectancy_summary()`, with INE's published values.

	`inedemogR`		INE	
Province	e_0	e_{65}	e_0	e_{65}
Madrid	87.86	24.75	87.74	24.62
Barcelona	86.87	23.86	86.84	23.82
Sevilla	85.38	22.45	85.35	22.42
A Coruña	86.95	23.83	86.88	23.76

Table 4
Differences between `inedemogR` and INE (years), 2002–2024, by INE's closing age group.

Provinces	Measure	n	Mean signed	Mean abs.	Max abs.	Within 0.25
50 (INE closes at 95+)	e_0	3450	0.010	0.031	0.202	100%
	e_{65}	3450	0.012	0.035	0.211	100%
Ceuta, Melilla (90+)	e_0	138	0.148	0.182	0.674	70%
	e_{65}	138	0.169	0.206	0.758	67%

Fig. 2(b) maps female e_0 for every province, and Table 3 sets four of them beside INE's published values. For A Coruña in 2024, male e_0 is 81.21 years, and the independently built abridged table gives a female e_0 of 86.95. Because both tables use the same inputs, their agreement is a consistency check only; the external benchmark follows.

3.3. Validation against INE's provincial life tables

INE's provincial tables [13] estimate the same quantity from death microdata. They use exact exposure times, observed a_x at every age and no old-age smoothing, and they are published in five-year groups closing at 95+, or at 90+ for Ceuta and Melilla. We compared e_0 and e_{65} for all 52 provinces, both sexes and the total, 2002–2024 (3,588 comparisons per measure), against a tolerance of 0.25 years fixed before the comparison (Table 4).

For the 50 provinces whose INE tables close at 95+, every comparison is within tolerance. An Arriaga decomposition of each difference puts it almost entirely at ages 80+ (mean absolute contribution 0.038 years); ages 1–79 contribute 0.008 and age 0 contributes less than 0.002, so the two a_0 conventions do not matter. The inputs are not the source either: across 39,939 grouped death rates (ages 1–89) the median relative difference from INE's is at most 0.1% in absolute value, and 95% of rates based on 100 or more deaths agree within 1.9%. Building our

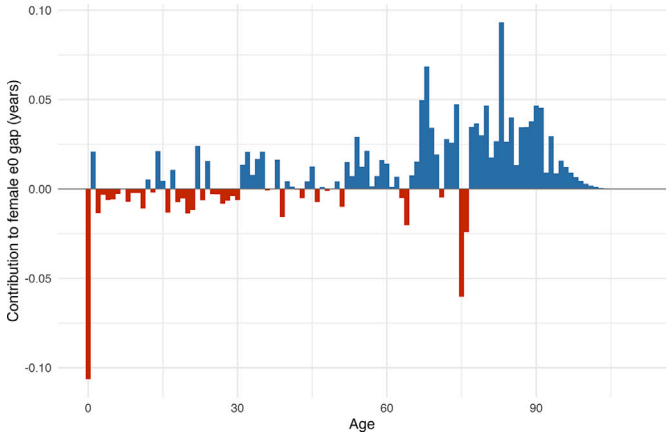


Fig. 3. Arriaga decomposition of the 2024 female e_0 gap between A Coruña and Madrid (0.91 years). Bars above zero favour Madrid. Both tables use Kannisto rates from the replacement age $Y = 95$; the contribution of ages 80+ also includes observed rates at 80–94.

tables without smoothing did not bring them closer to INE's. The remaining differences are concentrated at older ages and are consistent with differences in terminal-age treatment, particularly in Ceuta and Melilla, where INE closes its tables at 90+ with the observed mean time lived by that year's decedents: our e_{90} exceeds INE's by 0.78 years in Ceuta and 0.87 in Melilla on average. These comparisons do not establish which closure is more accurate, and agreement with INE is weaker for these two cities.

Sensitivity. Fixing Y at 80, 85, 90 or 95 instead of the 100-death rule changes e_0 by at most 0.05 years on average (0.32 at most, in provinces under 200,000 inhabitants). Starting the fit at 85 instead of 80 changes e_0 by 0.19 years on average in small provinces and up to 1.16, which supports the Protocol's 80. Moving the lower-triangle share of deaths from 0.5 to 0.45 or 0.55, or to 0.85 at age 0, changes e_0 and e_{65} by at most 0.014 years. These maxima refer to valid tables: in two small-province years the 0.45 share gave negative exposure at the oldest ages, and those tables were withheld. A Poisson bootstrap of 2024 deaths (200 replicates) conditions on the exposures and modelling choices, so it measures death-count variability only. The median bootstrap standard deviation of e_0 is 0.14 years for women and 0.16 for men in provinces over one million inhabitants, and 0.43 and 0.46 years in those under 200,000 (0.77 at most). Differences of a few tenths of a year between small provinces are of the same order as this variability; a contrast between two particular provinces needs its own uncertainty.

3.4. Life-expectancy decomposition and the mortality surface

```
female <- lt$fltper                    # female tables only
lt_a <- subset(female, province_name == "A Coruna" & year == 2024)
lt_b <- subset(female, province_name == "Madrid"   & year == 2024)
d <- decompose_life_expectancy(lt_a, lt_b, method = "arriaga")
attr(d, "residual")                     # 0 for Arriaga
plot_lexis_diagram(rates$mx_1x1, province = "Madrid")
```

The 2024 female e_0 gap between A Coruña and Madrid is 0.91 years (Fig. 3). A Coruña has lower mortality at age 0 and in the mid-seventies; Madrid's lead comes from the late sixties and above all from ages 80+ (0.57 years; 95% bootstrap interval 0.44–0.71). The gap itself is uncertain (0.64–1.22), and ages 0–64 contribute to an interval that includes zero. Pollard's contributions sum to the gap within 0.001 years. Between 2006 and 2024, female e_0 in A Coruña rose by 2.85 years. Fig. 4 shows Madrid's mortality surface; the 2020 rise at older ages appears as a vertical band.

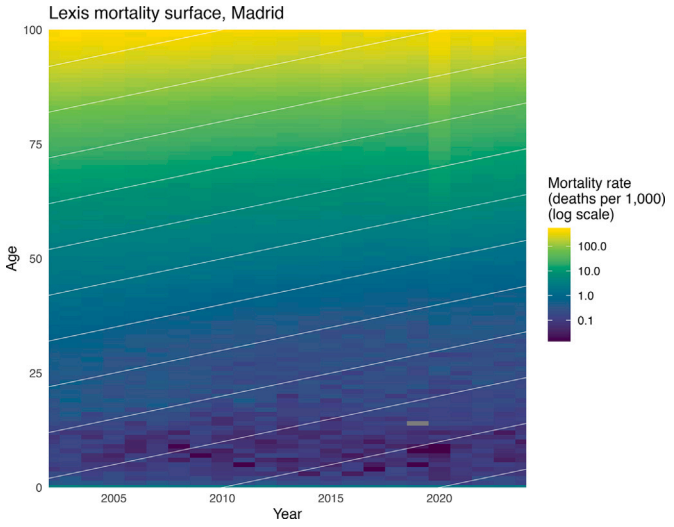


Fig. 4. Lexis surface of $m(x,t)$ for Madrid, both sexes, 2002–2024 (`plot_lexis_diagram`), shown in deaths per 1000 on a $\log_{10}$ colour scale; diagonals mark birth cohorts. A grey cell indicates an observed zero rate, which a log scale cannot show (the only one is at age 14 in 2019), not missing data. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.5. Fertility

```
births_age <- get_ine_births_by_age(n_periods = 100, cache_dir = cache)
asfr <- age_specific_fertility_rate(births_age$data, pop$data)
tfr  <- total_fertility_rate(asfr)        # children per woman
mac  <- mean_age_at_childbearing(asfr)
nrr  <- net_reproduction_rate(asfr, lt$fltper)
```

In 2024 the TFR is 0.98 in A Coruña, 1.05 in Barcelona, 1.09 in Madrid and 1.21 in Sevilla, and the mean age at childbearing is 32.5–33.5 years (Fig. 5). `asfr` is in births per woman; `asfr_per_1000` is provided for plotting. Against INE's published provincial series [14] for all 52 provinces, 1975–2024 (2,600 province-years), the mean absolute difference is 0.003 for both TFR (children per woman) and MAC (years), with maxima of 0.017 and 0.020; 99% of TFR and 98% of MAC values agree within 0.01, the precision INE publishes. Under the conventions of version 0.1.0, MAC was about 0.55 years too low.

3.6. Population structure and natural change

```
plot_population_pyramid(pop$data, year = 2024, region = "Sevilla")
cbr <- crude_birth_rate(get_ine_births(cache_dir = cache)$data, pop$data)
rni <- rate_of_natural_increase(cbr, crude_death_rate(deaths$data_provinces,
                                                      pop$data))
```

The pyramids (Fig. 6) show the baby-boom bulge and a narrow base. Natural increase has been negative in A Coruña throughout 2002–2024, in Barcelona since 2018 and in Sevilla since 2020; Madrid turned negative in 2020 and was slightly positive again in 2024.

4. Impact

`inedemogR` replaces the manual work between INE's public API and a provincial life table with a short script that a third party can rerun, and every reported result can be regenerated from the frozen inputs in the repository. Because its results agree with INE's own tables (Section 3.3) and its departures from the HMD protocol are documented, the workflow can support research on how provincial mortality differs and converges over time; `decompose_life_expectancy` returns the age structure of any such gap directly. It also suits teaching, since every intermediate quantity of the HMD protocol is a tidy table.

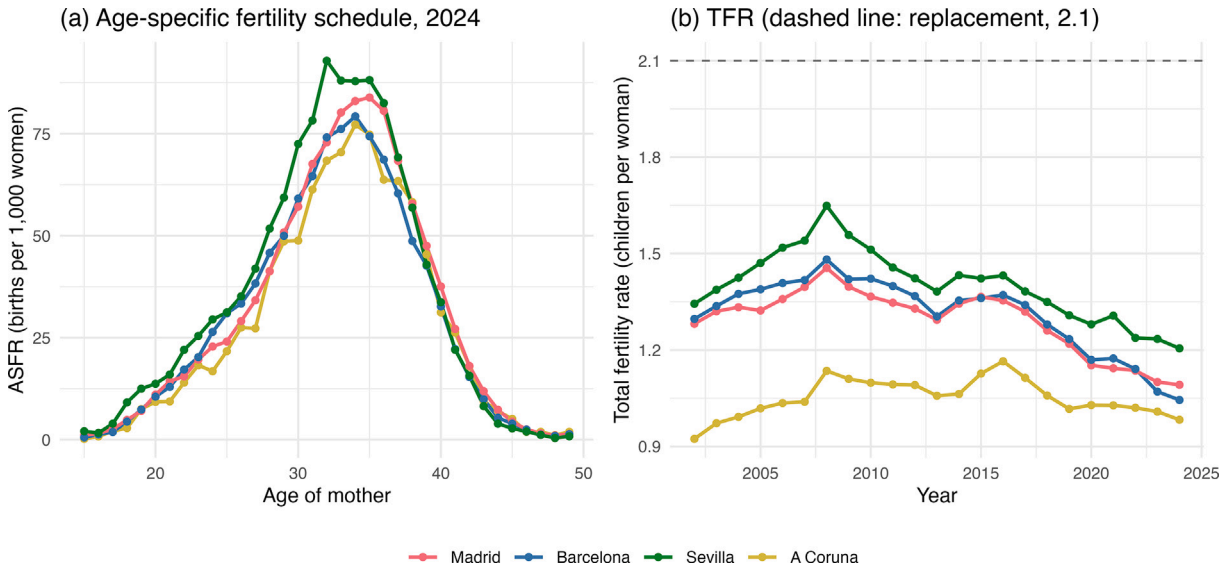


Fig. 5. Madrid, Barcelona, Sevilla and A Coruña. (a) Age-specific fertility, 2024, in births per 1000 women (`asfr_per_1000`; the function returns births per woman); (b) Total fertility rate, 2002–2024, in children per woman, with the replacement level of 2.1.

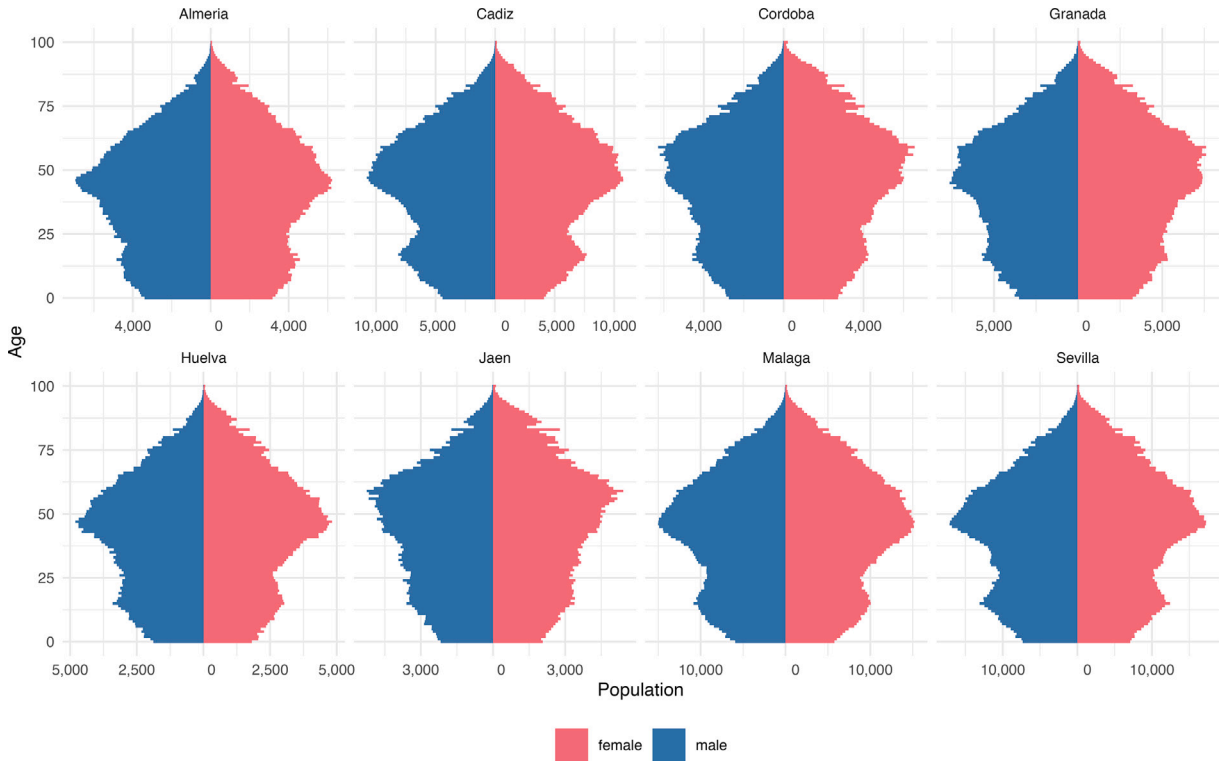


Fig. 6. Population pyramids, Andalusian provinces, 1 January 2024; each panel is on its own scale.

The design has two audiences. Users who model elsewhere can stop at the cleaned data, exported as CSV or HMD-format files. Users who stay in R can apply the analysis functions to the same objects, or to data from another source in the documented shape. The retrieval layer is specific to INE; the analytical core is not, and another statistical office's API could be connected by rewriting only that layer.

Testing against live data has shaped the package: it has uncovered defects that synthetic fixtures had not shown, and the review of version 0.1.0 has uncovered more (Section 2.3). Each has a regression test. The package has been on CRAN since August 2026. The author uses it in ongoing work comparing HMD-protocol and national life-table methods for small provinces.

5. Conclusions

`inedemogR` carries INE's demographic data from the public API to analysis-ready tables and provincial life tables in one documented workflow. Data-integrity checks at every stage and deterministic tests of the calculations support its correctness internally. Externally, its results agree with INE's published provincial series: within 0.25 years in e_0 and

e_{65} for all provinces whose life tables close at 95+, less closely in Ceuta and Melilla, whose INE tables close at 90+, and within 0.01 in 99% of TFR and 98% of MAC values. The current scope has limits. Life tables cover 2002 onwards, because earlier INE populations are top-coded at 85+. They are period tables, not cohort tables. Exposure uses an even Lexis-triangle split, and old-age mortality is extrapolated from ages 80–99. In small provinces the bootstrap standard deviation of e_0, which reflects only death-count variability, reaches about 0.8 years. The age- and sex-specific detail needed for life tables comes from province-level tables; municipal totals are available through the quick-lookup front end only. International migration has not yet been retrieved. Results depend on INE's revisable series, so published analyses should keep the provenance records of the inputs. Planned work adds cohort-component projection, nuptiality indicators and age-heaping diagnostics.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No new data were created. The INE inputs used in this article (retrieved on 5 October 2026), INE's reference life tables, and the scripts that regenerate every reported number, table and figure are in the `reproducibility/` folder of the code repository (https://github.com/jrcarob/inedemogR_package); the package itself is released on CRAN [15]. All INE data are public at https://www.ine.es.

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